Band-1 Hadamard Order 5924 — Miyamoto (Double Paley-I H_1480 + C_1482 over GF(1481)); IN SPK via Paley-I; Gap 2052 Still OPEN

H_740/H_1480/C_1482/H_5924 Miyamoto PASS; H_5924 Paley-I PASS (SPK); Paley-II SKIP; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_5812_Miyamoto`; `2026-09-15_Band1_Hadamard_5716_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_5924_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 740, 1480 and 5924 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double Paley-I seed H_{1480} = Sylvester-double(H_{740}) where H_{740} = Paley-I($q = 739$), and symmetric Paley conference C_{1482} over **GF(**1481**)** (prime field, $q = 1481 \equiv 1 \pmod 4$). Paley-I direct for H_{1480} is **SKIP** ($q = 1479 = 3 \cdot 17 \cdot 29$ not prime). Classical SPK path **PASS**: Paley-I ($q = 5923$ prime $\equiv 3 \pmod 4$, verified $\max|HH^\top - nI| = 0$). Paley-II **SKIP** ($q = 2961 = 3^2 \cdot 7 \cdot 47$ not prime); Kronecker **FAIL** (4×1481). Therefore order **5924 is IN SPK via Paley-I** (Miyamoto is the constructive seed path for the paper, not the SPK membership claim). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. `clay_closed = false`.

Abstract

Finite constructions of orders **740**, **1480**, and **5924**.

- H_{740} = Paley-I ($q = 739$)
- H_{1480} = Sylvester double of H_{740} (Paley-I direct SKIP: $q = 1479$ not prime)
- C_{1482} = Paley conference over GF(1481) prime field
- H_{5924} = Miyamoto(H_{1480}, C_{1482})
- H_{5924} Paley-I ($q = 5923$) — **SPK path PASS**
- Paley-II for 5924: **SKIP**

SPK: 5924 IN SPK via Paley-I (not Miyamoto-only). Seeds 740/1480 enter SPK via Paley-I / Double.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_5924.py`

Results. overall_pass=true; units 72/72; wall 358.0289s; all max|HH −nI|=0.

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Arithmetic

- Target $n = 5924 = 4 \cdot 1481$; 1481 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1480} and C_{1482} (Paley conference over prime field 1481).
- Seed: Double of Paley-I H_{740} ($q = 739$ prime $\equiv 3 \pmod 4$).
- Paley-I direct for 1480 SKIP: $q = 1479$ not prime.
- SPK: Paley-I $q = 5923$ **PASS**; Paley-II $q = 2961$ **SKIP**; Kronecker **FAIL** → **IN SPK via Paley-I**.

Constructions (verified)

Object	Method	max|HH −nI|	Wall (s)
H_740 Paley-I	Paley-I q=739	0	0.4887
H_1480 Paley-I	SKIP q=1479 not prime	—	—
H_1480 Double	Sylvester double of H_740	0	1.4529
C_1482	Paley conf GF(1481) q=1481	0	2.513
H_5924 Miyamoto	Miyamoto(H_1480, C_1482)	0	164.7794
H_5924 Paley-I	Paley-I q=5923 (SPK)	0	183.4963
H_5924 Paley-II	SKIP q=2961 not prime	—	—

SPK closure note

Seeds include prior SPK ships through **5812**'s seed **1452** + verified **740/1480/5924** (Paley-I); **exclude** Miyamoto-only bonuses including **5812/5476/5284**. Known count 1232. Next constructive gap **2052** (OPEN).

Quantity	Value
5924 in SPK	**True** (via Paley-I)
1480 in SPK	**True** (Double Paley-I)
740 in SPK	**True** (Paley-I)
5812 in SPK	**False** (prior Miyamoto-only)
5716 in SPK	**True** (prior Paley-II)
Next constructive gap	**2052** (OPEN)

Honesty

clay_closed = false · GATE-04 LOCKED · Hadamard conjecture OPEN · gap 2052 OPEN · no RH/P NP/YM/Frankl/NS/AC/Clay-closed claims.